

Prashant Sharma

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Education

International Institute of Information Technology Bangalore

M.Tech in Computer Science and Engineering

July 2025 – Present

CGPA: 3.21 / 4.0

GB Pant DSEU, Okhla Campus

Bachelor of Technology in Computer Science and Engineering

July 2021 – June 2025

CGPA: 3.27 / 4.0

Army Public School, Delhi Cantt

Higher Secondary Examination (CBSE)

April 2019 – June 2020

Percentage: 93.6

Projects

Visual Product Search Engine

YOLO, BLIP-2, CLIP, Streamlit, PyTorch

GitHub

- Built a **multimodal fashion retrieval pipeline** using **YOLO** for localization, **BLIP-2** for caption generation, and **fine-tuned CLIP** for cross-modal embeddings on the DeepFashion dataset.
- Applied **HNSW vector indexing** with fused image-text embeddings and **BLIP-2 ITM re-ranking** to improve retrieval relevance.
- Achieved **Recall@10 of 0.7396** and **NDCG@10 of 0.5070**; deployed on **Streamlit Community Cloud** with FAISS search and Hugging Face hosting.

Mental Health Score Prediction Platform

Python, Scikit-learn, Pydantic, Render

GitHub

- Developed an end-to-end machine learning platform to predict a continuous **mental health score (0–10)** using a **Random Forest Regressor** trained on 12 behavioral and lifestyle features.
- Implemented **FastAPI REST APIs** and integrated them with a responsive HTML/CSS/JavaScript frontend, enabling real-time prediction through a cloud-hosted web application.
- Achieved an R^2 **score of 0.86** with an **MAE of 0.35**, and deployed the application on **Render** for scalable online access.

BYOM Model Serving System

FastAPI, PostgreSQL, SQLAlchemy, JWT

GitHub

- Developed a **multi-user ML model** serving platform enabling secure user registration, authentication, model upload (**.pkl/.joblib**), and real-time inference through **10+ REST APIs** supporting end-to-end model lifecycle management.
- Implemented secure multi-tenant model serving with role-based access control. Supported concurrent storage and inference for **50+ uploaded models** during testing..
- Implemented multi-tenant platform capabilities including **user-specific prediction quotas**, **admin quota management**, **automatic artifact cleanup**, and **isolated model storage**, ensuring secure serving of models across multiple users

Technical Skills

Programming & Tools: Python, MySQL, FastAPI, Git, Docker

Data Science & ML: NumPy, Pandas, Scikit-learn, XGBoost, Seaborn, Matplotlib

Deep Learning & NLP: PyTorch, Pydantic, Word2Vec, HuggingFace Transformers

Coursework: Data Structures & Algorithms, Machine Learning, Visual Recognition, NLP, Databases, Operating System

Achievements

- Qualified **GATE CSE 2025** with **98.7 percentile**.
- Won Gold medal at **International Maths Olympiad, 2020** at School level.
- National Service Scheme (NSS) Member** July 2022 – June 2023
 - Led community service initiatives including environmental drives, eco-events, and blood donation campaigns.